Everyday Math

154. **Buying by the case** Joe can buy his favorite ice tea at a convenience store for $\$1.99$ per bottle. At the grocery store, he can buy a case of 12 bottles for $\$23.88$.
ⓐ Use the distributive property to find the cost of 12 bottles bought individually at the convenience store. (Hint: notice that $\$1.99$ is $\$2 - \$0.01$.)

ⓑ Is it a bargain to buy the iced tea at the grocery store by the case?

155. **Multi-pack purchase** Adele's shampoo sells for $\$3.97$ per bottle at the drug store. At the warehouse store, the same shampoo is sold as a 3-pack for $\$10.49$.
ⓐ Show how you can use the distributive property to find the cost of 3 bottles bought individually at the drug store.

ⓑ How much would Adele save by buying the 3-pack at the warehouse store?

Writing Exercises

156. Simplify $8\left(x - \frac{1}{4}\right)$ using the distributive property and explain each step.

157. Explain how you can multiply $4(\$5.97)$ without paper or a calculator by thinking of $\$5.97$ as $6 - 0.03$ and then using the distributive property.

Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
simplify expressions using the Distributive Property.			
evaluate expressions using the Distributive Property.			

ⓑ *What does this checklist tell you about your mastery of this section? What steps will you take to improve?*

7.4 Properties of Identity, Inverses, and Zero

Learning Objectives

By the end of this section, you will be able to:

- Recognize the identity properties of addition and multiplication
- Use the inverse properties of addition and multiplication
- Use the properties of zero
- Simplify expressions using the properties of identities, inverses, and zero

☑ **BE PREPARED** 7.10 Before you get started, take this readiness quiz.

Find the opposite of -4.
If you missed this problem, review Example 3.3.

☑ **BE PREPARED** 7.11 Find the reciprocal of $\frac{5}{2}$.
If you missed this problem, review Example 4.29.

☑ **BE PREPARED** 7.12 Multiply: $\frac{3a}{5} \cdot \frac{9}{2a}$.
If you missed this problem, review Example 4.27.

Recognize the Identity Properties of Addition and Multiplication

What happens when we add zero to any number? Adding zero doesn't change the value. For this reason, we call 0 the **additive identity**.

For example,

$13+0$	$-14+0$	$0+(-3x)$
13	-14	$-3x$

What happens when you multiply any number by one? Multiplying by one doesn't change the value. So we call 1 the **multiplicative identity**.

For example,

$43 \cdot 1$	$-27 \cdot 1$	$1 \cdot \frac{6y}{5}$
43	-27	$\frac{6y}{5}$

Identity Properties

The **identity property of addition**: for any real number a,

$$a+0=a \qquad 0+a=a$$

0 is called the **additive identity**

The **identity property of multiplication**: for any real number a

$$a \cdot 1=a \qquad 1 \cdot a=a$$

1 is called the **multiplicative identity**

EXAMPLE 7.33

Identify whether each equation demonstrates the identity property of addition or multiplication.

ⓐ $7+0=7$ ⓑ $-16(1)=-16$

Solution

ⓐ

	$7+0=7$
We are adding 0.	We are using the identity property of addition.

ⓑ

	$-16(1)=-16$
We are multiplying by 1.	We are using the identity property of multiplication.

TRY IT 7.65 Identify whether each equation demonstrates the identity property of addition or multiplication:

ⓐ $23+0=23$ ⓑ $-37(1)=-37$.

TRY IT 7.66 Identify whether each equation demonstrates the identity property of addition or multiplication:

ⓐ $1 \cdot 29=29$ ⓑ $14+0=14$.

Use the Inverse Properties of Addition and Multiplication

What number added to 5 gives the additive identity, 0?	
$5+$ _____ $= 0$	We know $5 + (-5) = 0$
What number added to -6 gives the additive identity, 0?	
$-6+$ _____ $= 0$	We know $-6 + 6 = 0$

Notice that in each case, the missing number was the opposite of the number.

We call $-a$ the **additive inverse** of a. The opposite of a number is its additive inverse. A number and its opposite add to 0, which is the additive identity.

What number multiplied by $\frac{2}{3}$ gives the multiplicative identity, 1? In other words, two-thirds times what results in 1?

$\frac{2}{3} \cdot$ ___ $= 1$	We know $\frac{2}{3} \cdot \frac{3}{2} = 1$

What number multiplied by 2 gives the multiplicative identity, 1? In other words two times what results in 1?

$2 \cdot$ ___ $= 1$	We know $2 \cdot \frac{1}{2} = 1$

Notice that in each case, the missing number was the reciprocal of the number.

We call $\frac{1}{a}$ the **multiplicative inverse** of $a (a \neq 0)$. The reciprocal of a number is its multiplicative inverse. A number and its reciprocal multiply to 1, which is the multiplicative identity.

We'll formally state the Inverse Properties here:

Inverse Properties

Inverse Property of Addition for any real number a,

$$a + (-a) = 0$$

$-a$ is the **additive inverse** of a.

Inverse Property of Multiplication for any real number $a \neq 0$,

$$a \cdot \frac{1}{a} = 1$$

$\frac{1}{a}$ is the **multiplicative inverse** of a.

EXAMPLE 7.34

Find the additive inverse of each expression: ⓐ 13 ⓑ $-\frac{5}{8}$ ⓒ 0.6.

Solution

To find the additive inverse, we find the opposite.

ⓐ The additive inverse of 13 is its opposite, -13. ⓑ The additive inverse of $-\frac{5}{8}$ is its opposite, $\frac{5}{8}$.

ⓒ The additive inverse of 0.6 is its opposite, -0.6.

TRY IT 7.67 Find the additive inverse: ⓐ 18 ⓑ $\frac{7}{9}$ ⓒ 1.2.

TRY IT 7.68 Find the additive inverse: ⓐ 47 ⓑ $\frac{7}{13}$ ⓒ 8.4.

EXAMPLE 7.35

Find the multiplicative inverse: ⓐ 9 ⓑ $-\frac{1}{9}$ ⓒ 0.9.

Solution

To find the multiplicative inverse, we find the reciprocal.

ⓐ The multiplicative inverse of 9 is its reciprocal, $\frac{1}{9}$. ⓑ The multiplicative inverse of $-\frac{1}{9}$ is its reciprocal, -9.

ⓒ To find the multiplicative inverse of 0.9, we first convert 0.9 to a fraction, $\frac{9}{10}$. Then we find the reciprocal, $\frac{10}{9}$.

TRY IT 7.69 Find the multiplicative inverse: ⓐ 5 ⓑ $-\frac{1}{7}$ ⓒ 0.3.

TRY IT 7.70 Find the multiplicative inverse: ⓐ 18 ⓑ $-\frac{4}{5}$ ⓒ 0.6.

Use the Properties of Zero

We have already learned that zero is the additive identity, since it can be added to any number without changing the number's identity. But zero also has some special properties when it comes to multiplication and division.

Multiplication by Zero

What happens when you multiply a number by 0? Multiplying by 0 makes the product equal zero. The product of any real number and 0 is 0.

Multiplication by Zero

For any real number a,

$$a \cdot 0 = 0 \qquad 0 \cdot a = 0$$

EXAMPLE 7.36

Simplify: ⓐ $-8 \cdot 0$ ⓑ $\frac{5}{12} \cdot 0$ ⓒ $0(2.94)$.

Solution

ⓐ

	$-8 \cdot 0$
The product of any real number and 0 is 0.	0

ⓑ

	$\frac{5}{12} \cdot 0$
The product of any real number and 0 is 0.	0

ⓒ

	$0(2.94)$
The product of any real number and 0 is 0.	0

TRY IT 7.71 Simplify: ⓐ $-14 \cdot 0$ ⓑ $0 \cdot \frac{2}{3}$ ⓒ $(16.5) \cdot 0$.

TRY IT 7.72 Simplify: ⓐ $(1.95) \cdot 0$ ⓑ $0(-17)$ ⓒ $0 \cdot \frac{5}{4}$.

Dividing with Zero

What about dividing with 0? Think about a real example: if there are no cookies in the cookie jar and three people want to share them, how many cookies would each person get? There are 0 cookies to share, so each person gets 0 cookies.

$$0 \div 3 = 0$$

Remember that we can always check division with the related multiplication fact. So, we know that

$$0 \div 3 = 0 \text{ because } 0 \cdot 3 = 0.$$

Division of Zero

For any real number a, except 0, $\frac{0}{a} = 0$ and $0 \div a = 0$.

Zero divided by any real number except zero is zero.

EXAMPLE 7.37

Simplify: ⓐ $0 \div 5$ ⓑ $\frac{0}{-2}$ ⓒ $0 \div \frac{7}{8}$.

Solution

ⓐ

	$0 \div 5$
Zero divided by any real number, except 0, is zero.	0

ⓑ

	$\frac{0}{-2}$
Zero divided by any real number, except 0, is zero.	0

ⓒ

	$0 \div \frac{7}{8}$
Zero divided by any real number, except 0, is zero.	0

TRY IT 7.73 Simplify: ⓐ $0 \div 11$ ⓑ $\frac{0}{-6}$ ⓒ $0 \div \frac{3}{10}$.

TRY IT 7.74 Simplify: ⓐ $0 \div \frac{8}{3}$ ⓑ $0 \div (-10)$ ⓒ $0 \div 12.75$.

Now let's think about dividing a number *by* zero. What is the result of dividing 4 by 0? Think about the related multiplication fact. Is there a number that multiplied by 0 gives 4?

$$4 \div 0 = ___ \text{ means } ___ \cdot 0 = 4$$

Since any real number multiplied by 0 equals 0, there is no real number that can be multiplied by 0 to obtain 4. We can conclude that there is no answer to $4 \div 0$, and so we say that division by zero is undefined.

Division by Zero

For any real number a, $\frac{a}{0}$, and $a \div 0$ are undefined.

Division *by* zero is undefined.

EXAMPLE 7.38

Simplify: ⓐ $7.5 \div 0$ ⓑ $\frac{-32}{0}$ ⓒ $\frac{4}{9} \div 0$.

Solution

ⓐ

	$7.5 \div 0$
Division by zero is undefined.	undefined

ⓑ

	$\frac{-32}{0}$
Division by zero is undefined.	undefined

ⓒ

	$\frac{4}{9} \div 0$
Division by zero is undefined.	undefined

TRY IT 7.75 Simplify: ⓐ $16.4 \div 0$ ⓑ $\frac{-2}{0}$ ⓒ $\frac{1}{5} \div 0$.

TRY IT 7.76 Simplify: ⓐ $\frac{-5}{0}$ ⓑ $96.9 \div 0$ ⓒ $\frac{4}{15} \div 0$

We summarize the properties of zero.

Properties of Zero

Multiplication by Zero: For any real number a,

$a \cdot 0 = 0 \qquad 0 \cdot a = 0$ The product of any number and 0 is 0.

Division by Zero: For any real number a, $a \neq 0$

$\frac{0}{a} = 0$ Zero divided by any real number, except itself, is zero.

$\frac{a}{0}$ is undefined. Division by zero is undefined.

Simplify Expressions using the Properties of Identities, Inverses, and Zero

We will now practice using the properties of identities, inverses, and zero to simplify expressions.

EXAMPLE 7.39

Simplify: $3x + 15 - 3x$.

Solution

	$3x + 15 - 3x$
Notice the additive inverses, $3x$ and $-3x$.	$0 + 15$
Add.	15

TRY IT 7.77 Simplify: $-12z + 9 + 12z$.

TRY IT 7.78 Simplify: $-25u - 18 + 25u$.

EXAMPLE 7.40

Simplify: $4\,(0.25q)$.

Solution

	$4\,(0.25q)$
Regroup, using the associative property.	$[4\,(0.25)]\,q$
Multiply.	$1.00q$
Simplify; 1 is the multiplicative identity.	q

TRY IT 7.79 Simplify: $2(0.5p)$.

TRY IT 7.80 Simplify: $25(0.04r)$.

EXAMPLE 7.41

Simplify: $\frac{0}{n+5}$, where $n \neq -5$.

Solution

	$\frac{0}{n+5}$
Zero divided by any real number except itself is zero.	0

TRY IT 7.81 Simplify: $\frac{0}{m+7}$, where $m \neq -7$.

TRY IT 7.82 Simplify: $\frac{0}{d-4}$, where $d \neq 4$.

EXAMPLE 7.42

Simplify: $\frac{10-3p}{0}$.

Solution

	$\frac{10-3p}{0}$
Division by zero is undefined.	undefined

TRY IT 7.83 Simplify: $\frac{18-6c}{0}$.

TRY IT 7.84 Simplify: $\frac{15-4q}{0}$.

EXAMPLE 7.43

Simplify: $\frac{3}{4} \cdot \frac{4}{3}(6x + 12)$.

Solution

We cannot combine the terms in parentheses, so we multiply the two fractions first.

	$\frac{3}{4} \cdot \frac{4}{3}(6x + 12)$
Multiply; the product of reciprocals is 1.	$1\,(6x + 12)$
Simplify by recognizing the multiplicative identity.	$6x + 12$

TRY IT 7.85 Simplify: $\frac{2}{5} \cdot \frac{5}{2}(20y + 50)$.

TRY IT 7.86 Simplify: $\frac{3}{8} \cdot \frac{8}{3}(12z + 16)$.

All the properties of real numbers we have used in this chapter are summarized in Table 7.1.

Property	Of Addition	Of Multiplication
Commutative Property		
If a and b are real numbers then...	$a + b = b + a$	$a \cdot b = b \cdot a$
Associative Property		
If a, b, and c are real numbers then...	$(a + b) + c = a + (b + c)$	$(a \cdot b) \cdot c = a \cdot (b \cdot c)$
Identity Property	0 is the additive identity	1 is the multiplicative identity
For any real number a,	$a + 0 = a$ $0 + a = a$	$a \cdot 1 = a$ $1 \cdot a = a$
Inverse Property	$-a$ is the additive inverse of a	$a, a \neq 0$ $1/a$ is the multiplicative inverse of a
For any real number a,	$a + (-a) = 0$	$a \cdot \frac{1}{a} = 1$
Distributive Property If a, b, c are real numbers, then $a(b + c) = ab + ac$		
Properties of Zero		

Table 7.1 Properties of Real Numbers

Property	Of Addition	Of Multiplication
For any real number a,	$a \cdot 0 = 0$ $0 \cdot a = 0$	
For any real number $a, a \neq 0$	$\frac{0}{a} = 0$ $\frac{a}{0}$ is undefined	

Table 7.1 Properties of Real Numbers

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Multiplying and Dividing Involving Zero (https://www.openstax.org/l/24MultDivZero)

SECTION 7.4 EXERCISES

Practice Makes Perfect

Recognize the Identity Properties of Addition and Multiplication

In the following exercises, identify whether each example is using the identity property of addition or multiplication.

158. $101 + 0 = 101$

159. $\frac{3}{5}(1) = \frac{3}{5}$

160. $-9 \cdot 1 = -9$

161. $0 + 64 = 64$

Use the Inverse Properties of Addition and Multiplication

In the following exercises, find the multiplicative inverse.

162. 8

163. 14

164. -17

165. -19

166. $\frac{7}{12}$

167. $\frac{8}{13}$

168. $-\frac{3}{10}$

169. $-\frac{5}{12}$

170. 0.8

171. 0.4

172. -0.2

173. -0.5

Use the Properties of Zero

In the following exercises, simplify using the properties of zero.

174. $48 \cdot 0$

175. $\frac{0}{6}$

176. $\frac{3}{0}$

177. $22 \cdot 0$

178. $0 \div \frac{11}{12}$

179. $\frac{6}{0}$

180. $\frac{0}{3}$

181. $0 \div \frac{7}{15}$

182. $0 \cdot \frac{8}{15}$

183. $(-3.14)(0)$

184. $5.72 \div 0$

185. $\frac{\frac{1}{10}}{0}$

Simplify Expressions using the Properties of Identities, Inverses, and Zero

In the following exercises, simplify using the properties of identities, inverses, and zero.

186. $19a + 44 - 19a$

187. $27c + 16 - 27c$

188. $38 + 11r - 38$

189. $92 + 31s - 92$

190. $10(0.1d)$

191. $100(0.01p)$

192. $5(0.6q)$

193. $40(0.05n)$

194. $\frac{0}{r+20}$, where $r \neq -20$

195. $\frac{0}{s+13}$, where $s \neq -13$

196. $\frac{0}{u-4.99}$, where $u \neq 4.99$

197. $\frac{0}{v-65.1}$, where $v \neq 65.1$

198. $0 \div \left(x - \frac{1}{2}\right)$, where $x \neq \frac{1}{2}$

199. $0 \div \left(y - \frac{1}{6}\right)$, where $y \neq \frac{1}{6}$

200. $\frac{32-5a}{0}$, where $32 - 5a \neq 0$

201. $\frac{28-9b}{0}$, where $28 - 9b \neq 0$

202. $\frac{2.1+0.4c}{0}$, where $2.1 + 0.4c \neq 0$

203. $\frac{1.75+9f}{0}$, where $1.75 + 9f \neq 0$

204. $\left(\frac{3}{4} + \frac{9}{10}m\right) \div 0$, where $\frac{3}{4} + \frac{9}{10}m \neq 0$

205. $\left(\frac{5}{16}n - \frac{3}{7}\right) \div 0$, where $\frac{5}{16}n - \frac{3}{7} \neq 0$

206. $\frac{9}{10} \cdot \frac{10}{9}(18p - 21)$

207. $\frac{5}{7} \cdot \frac{7}{5}(20q - 35)$

208. $15 \cdot \frac{3}{5}(4d + 10)$

209. $18 \cdot \frac{5}{6}(15h + 24)$

Everyday Math

210. **Insurance copayment** Carrie had to have 5 fillings done. Each filling cost $80. Her dental insurance required her to pay 20% of the cost. Calculate Carrie's cost

ⓐ by finding her copay for each filling, then finding her total cost for 5 fillings, and

ⓑ by multiplying $5\,(0.20)\,(80)$.

ⓒ Which of the Properties of Real Numbers did you use for part (b)?

211. **Cooking time** Helen bought a 24-pound turkey for her family's Thanksgiving dinner and wants to know what time to put the turkey in the oven. She wants to allow 20 minutes per pound cooking time.

ⓐ Calculate the length of time needed to roast the turkey by multiplying $24 \cdot 20$ to find the number of minutes and then multiplying the product by $\frac{1}{60}$ to convert minutes into hours.

ⓑ Multiply $24\left(20 \cdot \frac{1}{60}\right)$.

ⓒ Which of the Properties of Real Numbers allows you to multiply $24\left(20 \cdot \frac{1}{60}\right)$ instead of $(24 \cdot 20)\frac{1}{60}$?

Writing Exercises

212. In your own words, describe the difference between the additive inverse and the multiplicative inverse of a number.

213. How can the use of the properties of real numbers make it easier to simplify expressions?

Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
recognize the Identity Properties of Addition and Multiplication.			
use the Inverse Properties of Addition and Multiplication.			
use the Properties of Zero.			
simplify expressions using the properties of identities, inverses, and zero.			